

AI SOC Copilot Testing Report

Objective

The purpose of testing is to validate that the AI SOC Copilot correctly processes uploaded security logs, extracts Indicators of Compromise (IOCs), performs AI-driven threat analysis, maps threats to the MITRE ATT&CK framework, calculates severity and risk scores, and generates comprehensive incident reports.

Test Environment

Parameter	Value
Operating System	Windows 11
Backend	FastAPI
Frontend	React + Vite
AI Model	Google Gemini
Supported Formats	TXT, JSON, CSV
Browser	Google Chrome

Functional Test Cases

TC ID	Feature	Test Case	Expected Result	Status
TC-01	File Upload	Upload valid TXT log	File uploaded successfully	Pass
TC-02	File Upload	Upload valid JSON log	File uploaded successfully	Pass
TC-03	File Upload	Upload valid CSV log	File uploaded successfully	Pass

TC-04	File Validation	Upload unsupported file (.pdf)	Error message displayed	Pass
TC-05	File Validation	Upload empty file	Validation error shown	Pass
TC-06	Log Parsing	Parse uploaded log	Events extracted successfully	Pass
TC-07	IOC Extraction	Extract IP addresses	IPs displayed correctly	Pass
TC-08	IOC Extraction	Extract Domains	Domains displayed correctly	Pass
TC-09	IOC Extraction	Extract URLs	URLs displayed correctly	Pass
TC-10	IOC Extraction	Extract Email Addresses	Emails displayed correctly	Pass
TC-11	IOC Extraction	Extract File Hashes	Hashes displayed correctly	Pass
TC-12	IOC Extraction	Extract CVE IDs	CVEs displayed correctly	Pass
TC-13	Threat Analysis	Detect attack type	Attack type generated	Pass
TC-14	Severity	Generate severity level	Severity displayed	Pass
TC-15	Risk Score	Generate risk score	Risk score displayed	Pass
TC-16	Confidence	Generate confidence score	Confidence displayed	Pass
TC-17	MITRE Mapping	Display tactic & technique	Mapping generated	Pass
TC-18	Timeline	Generate incident timeline	Timeline displayed	Pass
TC-19	Containment	Generate containment actions	Recommendations displayed	Pass
TC-20	Recovery	Generate recovery steps	Recovery plan displayed	Pass
TC-21	Incident Report	Generate complete report	Report displayed	Pass
TC-22	Dashboard	Dashboard cards update	Data displayed correctly	Pass
TC-23	PDF Export	Export report	PDF generated successfully	Pass
TC-24	Session Reset	Start new analysis	Previous data cleared	Pass
TC-25	Performance	Analyze normal log	Response generated successfully	Pass

Testing Results Summary

Metric	Result
Total Cases	Test 25
Passed	25
Failed	0
Pass Rate	100%

Only use "100%" if your team actually executed and verified all these tests. If not, replace the numbers with your real results.

Accuracy Observations

During testing, the AI SOC Copilot demonstrated the following:

- Successfully processed uploaded security logs.
 - Correctly extracted Indicators of Compromise (IOCs).
 - Generated meaningful threat classifications.
 - Produced MITRE ATT&CK mappings consistent with detected attack behavior.
 - Generated structured incident reports with executive and technical summaries.
 - Displayed risk scores and severity levels for analyst prioritization.
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Observed Limitations

These are reasonable observations that match an AI-assisted system:

- AI output quality depends on the completeness of uploaded logs.
- Some uncommon attack scenarios may require analyst review.
- Confidence and risk scores are AI-generated and should support—not replace—human decision-making.

- Performance may vary depending on log size and complexity.
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Suggested Improvements

- Add support for additional log formats (e.g., EVTX, PCAP).
 - Integrate real-time SIEM log ingestion.
 - Connect with external Threat Intelligence feeds.
 - Enable analyst feedback to improve future AI responses.
 - Provide historical incident comparison and trend analysis.
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Conclusion

Testing demonstrated that the AI SOC Copilot successfully performs log ingestion, IOC extraction, AI-driven threat analysis, MITRE ATT&CK mapping, severity assessment, and incident report generation. The integrated workflow supports SOC analysts by automating repetitive analysis tasks while providing structured, actionable outputs.